

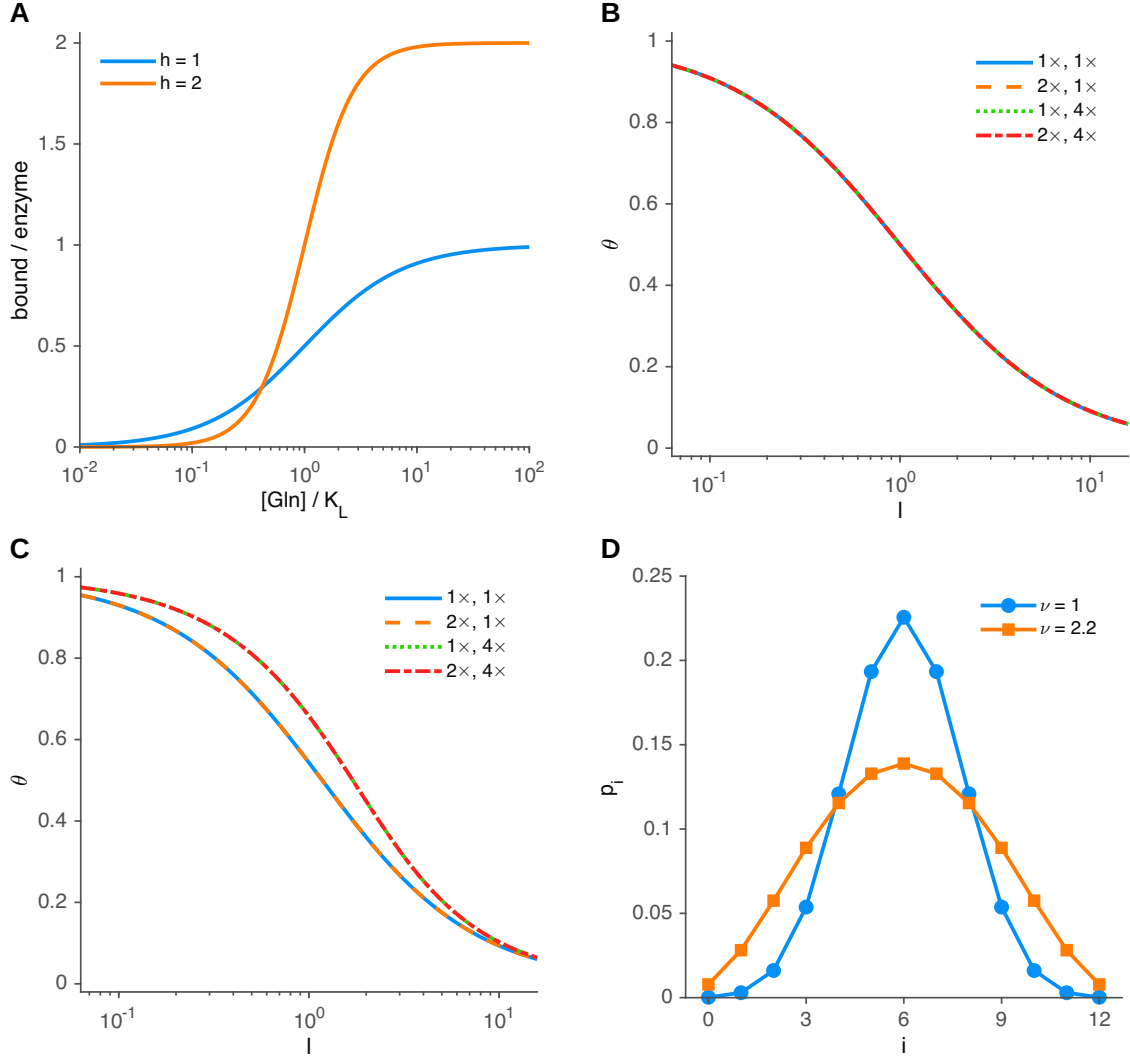


Figure S8: Measurements that discriminate unresolved mechanisms. (A) Illustrative one-ligand and concerted two-ligand binding laws, expressed as ligands per enzyme. Activity measurements must accompany occupancy to identify productive states. (B) Illustrative minimal-model total-target responses at two enzyme and two target levels. Parameters are $n=3$, $A=1$, $\hat{B}=\hat{G}=0.4$, $\hat{K}=1$, baseline $T_E=0.0001$ and $T_S=1$ in consistent concentration units; the four curves nearly coincide because the enzyme pool is dilute. (C) An illustrative active target-bridging extension with one modification site and relation $s_1=s_0/(1-0.35s_0)$, with total free target 1 or 4. Target levels separate; enzyme levels coincide in this reduced construction. These examples are not fits to the measured titration. (D) Twelve-site distributions with the same mean modification but $\nu=1$ or $\nu=2.2$, generated by exponential reweighting of binomial states as specified in S1 Appendix. These distributions are not fits to glutamine synthetase.